

S3.2 Functional groups: Classification of organic compounds

Functional groups

Learning outcomes

At the end of this section you should be able to identify the following functional groups by name and structure: halogeno, hydroxy (hydroxyl), carbonyl, carboxyl, alkoxy, amino, amido, ester, phenyl.

Organic compounds have many different uses but what makes them suitable for a particular use? Take, for example, the compounds ethanol and ethanoic acid. Both compounds are composed of carbon, hydrogen and oxygen atoms but have different properties and uses. Ethanol, for example, is used in hand sanitiser and ethanoic acid is a component of vinegar (**Figure 1**). What is different about these compounds? And what makes them suitable for their uses? In this section, you will learn why the chemical and physical properties differ between organic compounds.



Source: "COVID-19 au Bénin flacons de gel hydroalcoolique" [🔗](https://commons.wikimedia.org/wiki/File:COVID-19_au_B%C3%A9nin_flacons_de_gel_hydroalcoolique_sur_une_table1.jpg)

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sur une tablel  (https://commons.wikimedia.org/wiki/File:COVID-19_au_B%C3%A9nin_flacons_de_gel_hydroalcoolique_sur_une_tablel.jpg)” by Adoscam is licensed under [CC BY-SA 4.0](https://creativecommons.org/licenses/by-sa/4.0/deed.en)  (<https://creativecommons.org/licenses/by-sa/4.0/deed.en>)



Credit: Joe_Potato, Getty Images

Figure 1. Hand sanitiser and vinegar both contain organic compounds but have very different uses. What determines the physical and chemical properties of such compounds?

Organic compounds are organised into classes depending on the functional group that they contain. A functional group is an atom or group of atoms that give an organic compound its characteristic physical and chemical properties. Compounds with the same functional group are organised into the same class, and compounds in the same class have similar physical and chemical properties: these will be covered in more detail in the next section. In this section, you will learn how to identify the different functional groups present in organic compounds and also to determine to which class they belong.

Figure 2 shows three organic compounds. What do they have in common?

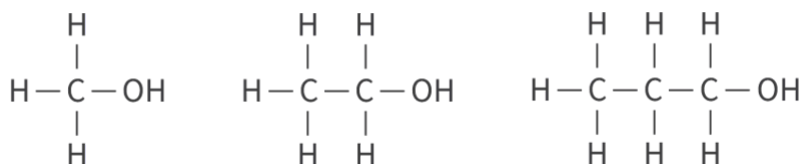


Figure 2. The full structural formulas of the first three members of the alcohol homologous series.

 More information for figure 2

The image displays the structural formulas of the first three members of the alcohol homologous series, arranged from left to right.

1. Methanol (CH₃OH):

2. One carbon atom (C) is centrally bonded to three hydrogen atoms (H) and one hydroxyl group (—OH).

3. Ethanol (C₂H₅OH):

4. Two carbon atoms are linked in a straight chain. The first carbon is bonded to three hydrogen atoms, and the second carbon is bonded to two hydrogen atoms and one hydroxyl group.

5. Propanol (C₃H₇OH):

6. Three carbon atoms are linked in a linear chain. The first two carbons are each bonded to two hydrogen atoms, and the last carbon is bonded to two hydrogen atoms and one hydroxyl group.

Each compound has an oxygen atom bonded to a hydrogen atom, represented as the hydroxyl functional group. This group is common to all alcohols, giving them similar chemical properties.

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They all have carbon and hydrogen atoms but then so do almost all organic compounds. The other common feature that they have is an oxygen atom bonded to a hydrogen atom: this is the functional group (—OH). The name of the functional group is a hydroxy (or hydroxyl) group and it is common to the class of

organic compounds called the alcohols. So, all alcohols contain a hydroxy group and it gives the class of compounds their characteristic physical and chemical properties.

Hydrocarbons

Alkanes, alkenes and alkynes are hydrocarbons: molecules composed of carbon and hydrogen only.

Alkanes are saturated: they have single covalent bonds between the carbon atoms.

Alkenes and alkynes are unsaturated: they have double bonds and triple bonds between the carbon atoms, respectively (**Figure 3**).

Double and triple covalent bonds were covered in section S2.2.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/>).

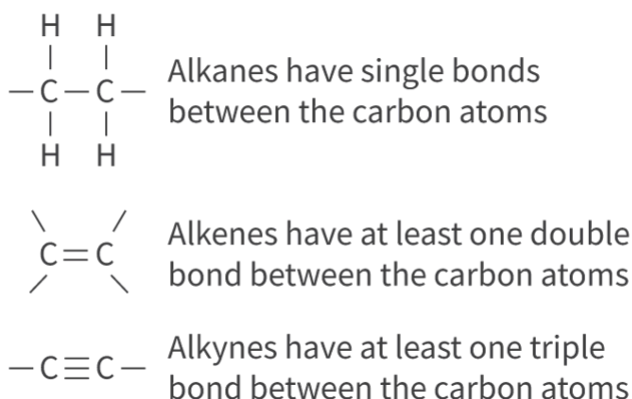


Figure 3. Alkanes, alkenes and alkynes have single, double and triple bonds between the carbon atoms, respectively.

 More information for figure 3

The image is a diagram illustrating the molecular structures of alkanes, alkenes, and alkynes. It shows the types of bonds between carbon atoms:

1. **Alkanes:** Depicted with single bonds between carbon atoms and represented by the molecular structure showing two carbon atoms connected by a single bond. Each carbon atom is bonded to two hydrogen atoms.

2. **Alkenes:** Displayed with at least one double bond between carbon atoms, represented by a structure showing two carbon atoms connected by a double bond and each carbon atom connected to one hydrogen atom.
3. **Alkynes:** Illustrated with at least one triple bond between carbon atoms, represented by two carbon atoms connected by a triple bond. Each carbon atom in the structure has one additional bond to a hydrogen atom.

The image helps in understanding the differences in chemical bonding and structure among these types of hydrocarbons.

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Figure 4 shows the ball and stick models of an alkane (ethane), an alkene (ethene) and an alkyne (ethyne). The naming of these molecules will be covered in section S3.2.5a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nomenclature-of-straight-chain-organic-id-45147/>).

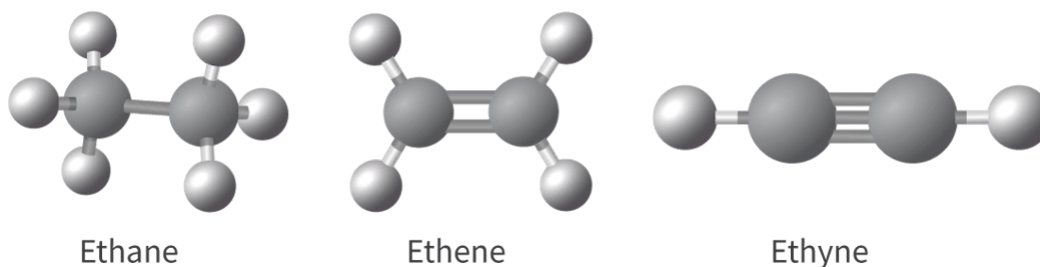


Figure 4. Ball and stick models and ethane, ethene and ethyne.

 More information for figure 4

The image shows ball and stick models for three types of hydrocarbon molecules: ethane, ethene, and ethyne. Each model represents the molecular structure using spheres for atoms and sticks for bonds.

1. **Ethane:** This model consists of two carbon atoms connected by a single bond, with each carbon atom also bonded to three hydrogen atoms. This structure represents an alkane, characterized by single bonds and a saturated carbon chain.

2. Ethene: This model includes two carbon atoms connected by a double bond. Each carbon atom is bonded to two hydrogen atoms. Ethene exemplifies an alkene, with a characteristic carbon-carbon double bond indicating unsaturation.
3. Ethyne: This model depicts two carbon atoms linked by a triple bond. Each carbon atom is connected to one hydrogen atom, showing an alkyne structure, which includes a carbon-carbon triple bond indicating a higher level of unsaturation compared to alkenes.

These models highlight the differences in bonding and structure among the hydrocarbon types.

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Making connections

How does the presence of double and triple bonds in molecules influence their reactivity (see [subtopic R2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43127/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43127/))?

Study skills

Throughout this section, you will see the letter R attached to the functional group. The letters R, R', etc represent alkyl groups. An alkyl group is a chain of carbon and hydrogen atoms only: think of it as an alkane with one hydrogen atom removed.

The names of the specific alkyl group are as follows:

- Methyl (—CH_3)
- Ethyl ($\text{—CH}_2\text{CH}_3$)
- Propyl ($\text{—CH}_2\text{CH}_2\text{CH}_3$)
- Butyl ($\text{—CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$)

In this section, the letter R does not actually represent any of the above groups. It is only used to represent the carbon chain part of the molecule.

Alcohols

Alcohols contain a hydroxy functional group, which is an oxygen atom bonded to a hydrogen atom by a single covalent bond (note that the bond between the O and the H is often omitted). **Figure 5** shows the hydroxy functional group along with methanol, the first alcohol in the series.

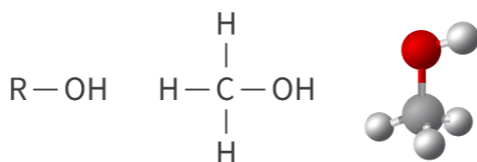


Figure 5. The hydroxy functional group (left) the full structural formula of methanol (centre) and the ball and stick model (right).

 More information for figure 5

The image depicts three representations. On the left is the hydroxy functional group represented as R-OH, where R is any alkyl group and OH denotes an oxygen single-bonded to a hydrogen. The center image shows the full structural formula for methanol, CH₃OH, with the carbon atom centered and bonded to three hydrogen atoms and one hydroxyl group. On the right is a ball-and-stick model of methanol, with a red sphere representing the oxygen atom, gray spheres for carbon and hydrogen, and sticks representing the covalent bonds.

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Aldehydes

Aldehydes contain a carbonyl functional group which is a carbon atom with a double bond to an oxygen atom (**Figure 6**). In the aldehydes, the carbonyl group is located in the terminal position (at one end of the molecule). This functional group is shared with other organic compounds, namely the ketones.

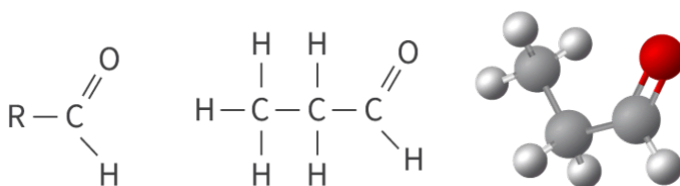


Figure 6. Aldehydes contain a carbonyl functional group (left). The full structural formula and ball and stick model of propanal (centre and right).

 More information for figure 6

The image consists of three parts representing the aldehyde propanal. On the left, a structural formula diagram shows a generic aldehyde functional group with symbols for atoms: 'R' attached to a carbon atom (C), double-bonded to an oxygen atom (O), and single-bonded to a hydrogen atom (H). The middle section depicts the expanded molecular structure of propanal, represented with letters and lines, showing all carbon (C) and hydrogen (H) atoms with a terminal carbonyl group. On the right, a ball-and-stick model displays a 3D representation of propanal, where spheres represent atoms connected by sticks standing for bonds; the carbonyl group shows a carbon (gray sphere) double-bonded to an oxygen (red sphere).

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Ketones

Like aldehydes, ketones also contain a carbonyl functional group. The difference between an aldehyde and a ketone is in an aldehyde the carbonyl group is located in the terminal position, but in ketones the carbonyl group is attached to a carbon atom within the carbon chain (**Figure 7**).

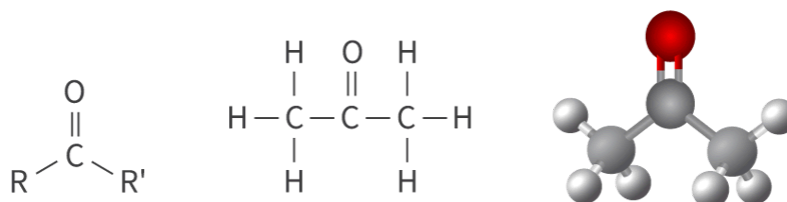


Figure 7. Ketones contain a carbonyl functional group (left). The full structural formula and ball and stick model of propanone (centre and left).

 More information for figure 7

The image consists of three representations related to ketones. On the left, a generic structural formula for a ketone is shown with a carbonyl group (C=O) bonded to two carbon groups labeled as R and R'. In the center, the full structural formula of propanone is depicted, consisting of three carbon atoms, with the middle carbon bonded to an oxygen atom twice (double-bonded). The hydrogens are bonded to the outer carbon atoms. On the right, a ball and stick model of propanone illustrates the same molecule, using spheres for atoms: gray for carbon, white for hydrogen, and red for oxygen, highlighting the carbonyl functional group in red at the top.

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Carboxylic acids

Carboxylic acids are organic acids that contain a carboxyl functional group, which is a combination of a carbonyl group and a hydroxyl group. A carboxyl functional group is a carbonyl group (C=O) and a hydroxyl group (OH) combined, represented as either COOH or CO₂H (**Figure 8**).

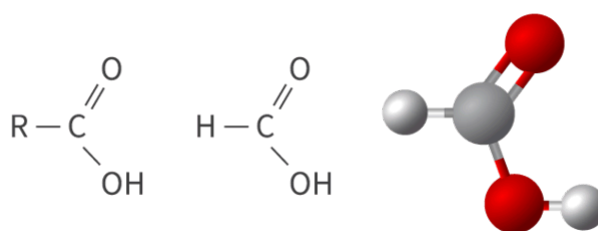


Figure 8. The carboxyl functional group (left) and the full structural formula of methanoic acid (centre) and the ball and stick model (right).

 More information for figure 8

The image shows the carboxyl functional group on the left, represented as R-C(=O)OH, where R represents a variable group. In the center, it's the structural formula of methanoic acid, H-C(=O)OH, illustrating how hydrogen (H) is attached to the carboxyl group. On the right, there's a ball and stick model of methanoic acid, depicting atoms as spheres connected with sticks to represent bonds: oxygen atoms are red, carbon is a larger grey sphere, and hydrogen appears as smaller white spheres. This model visualizes the three-dimensional arrangement of atoms in methanoic acid.

[Generated by AI]

Esters

Esters are a class of organic compounds known for their pleasant tastes and smells, which explains why they are used as artificial flavourings and in perfumes. The ester functional group is a carbonyl group with a single bond to another oxygen atom, and it is asymmetric. **Figure 9** shows the ester functional group, together with the ester ethyl ethanoate which has the distinctive smell of pear drops.

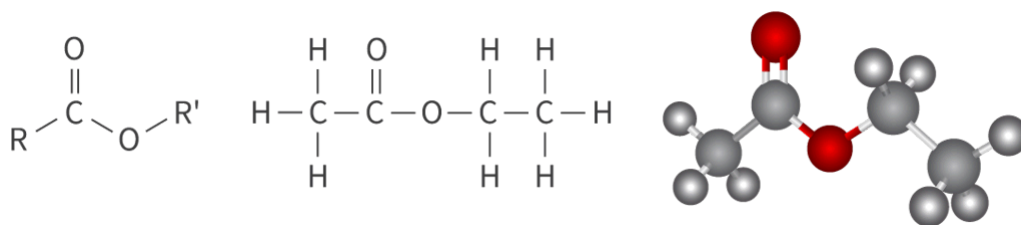


Figure 9. The ester functional group (left) and the full structural formula of ethyl ethanoate (centre) and the ball and stick model (right).

🔗 More information for figure 9

Ethers

Ethers have an alkoxy functional group that is composed of an oxygen atom bonded to two alkyl groups (**Figure 10**). Like the esters, ethers are also used in food flavourings and as fragrances in perfumes. Historically, one particular example, ethoxyethane, commonly referred to as ‘ether’, was used as an anaesthetic.

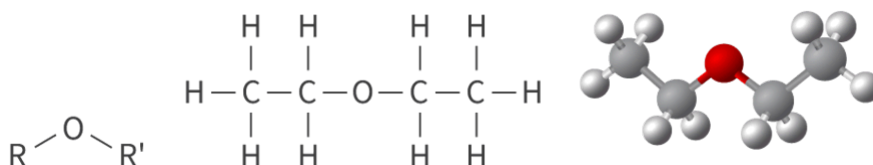


Figure 10. The ether functional group (left) and the full structural formula of ethoxyethane (centre) and the ball and stick model (right).

🔗 More information for figure 10

The image shows three representations related to ethers. On the left is the general formula for an ether functional group, depicted as $R-O-R'$, where R and R' are alkyl or aryl groups attached to the oxygen atom, forming an alkoxy group. In the center, the full structural formula of ethoxyethane is shown; it includes a chain with two ethyl groups (C_2H_5) on either side of an oxygen atom, represented as $H_3C-CH_2-O-CH_2-CH_3$. Finally, on the right, there is a ball and stick model illustrating the three-dimensional arrangement of atoms in ethoxyethane. The model uses grey balls for carbon atoms, white balls for hydrogen atoms, and a red ball for the oxygen atom, with sticks connecting them to show the bonds.

[Generated by AI]

Amines

Amines contain the amino functional group, which is a nitrogen atom bonded to hydrogen atoms or alkyl groups (**Figure 11**). Amines are derivatives of ammonia (NH_3) in which one or more of the hydrogen atoms has been replaced by an alkyl group. Amines can be classified as primary, secondary or tertiary depending on the number of substituents bonded to the nitrogen atom (this is covered in more detail in section S3.2.6 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structural-isomerism-id-45665/>)). It is worth noting that the nitrogen atom has a lone pair of electrons that is sometimes omitted in structural formulas.

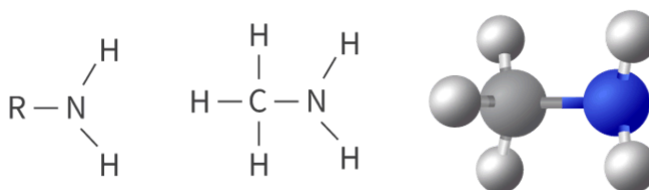


Figure 11. The amino functional group (left) and the full structural formula of methanamine (centre) and the ball and stick model (right).

 More information for figure 11

The image consists of three parts. On the left is the amino functional group, depicted as a nitrogen atom (N) bonded to two hydrogen atoms (H) and an unspecified group (R), indicated as R-N-H-H. In the center is the structural formula of methanamine, shown as H-C-H with each hydrogen atom bonded to the central carbon atom (C) and the carbon bonded to a nitrogen atom (N) which has one hydrogen bonded to it. To the right is a ball-and-stick model of methanamine, where atoms are represented as spheres and bonds as sticks. The carbon atom is a central larger sphere, with smaller spheres representing hydrogen atoms attached to it, and a nitrogen atom is visible with a distinctive color to indicate its difference from the carbon and hydrogen. This model provides a 3D perspective of the methanamine molecule, allowing visualization of atom positions and connectivity.

[Generated by AI]

Amides

Amides contain the amido functional group, which is a carbonyl group bonded to an amino group (**Figure 12**). Like the amines, they are also derivatives of ammonia and can be classified as either primary, secondary or tertiary.

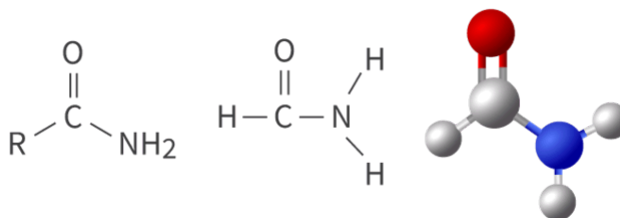


Figure 12. The amido functional group (left) and the full structural formula of methanamide (centre) and the ball and stick model (right).

 [More information for figure 12](#)

The image displays three diagrams related to amides. On the left, there is a structural representation of the amido functional group, characterized by a carbonyl group (a carbon double-bonded to an oxygen) attached to an amino group (a nitrogen bonded directly to the carbonyl carbon). In the center, the structural formula of methanamide (formamide) is shown, with its chemical components: the carbon (C) bonds with an oxygen (O) and a nitrogen (N) which is further bonded to two hydrogens (H). On the right, a ball-and-stick model illustrates the three-dimensional structure of methanamide, with atoms represented as spheres: the red sphere for oxygen, gray for carbon, blue for nitrogen, and white for hydrogen, demonstrating spatial arrangement.

[Generated by AI]

Halogenoalkanes

Halogenoalkanes are produced from alkanes in which one or more of the hydrogen atoms has been replaced with a halogen atom (either fluorine, chlorine, bromine or iodine, **Figure 13**). The name of the functional group is halogeno.

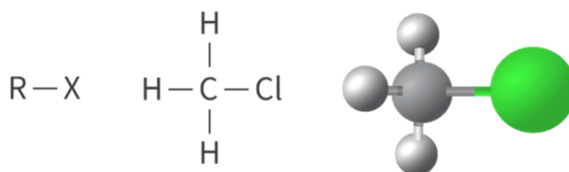


Figure 13. The halogeno functional group (left) and the full structural formula of chloromethane (centre) and the ball and stick model (right).

 [More information for figure 13](#)

The image is a diagram with three parts. On the left, there's a representation of a halogeno functional group labeled as 'R-X', where 'R' represents a generic alkyl group and 'X' a halogen. In the center, the structural formula of chloromethane is displayed, showing 'H-C-Cl' with two hydrogen atoms bonded to the carbon and a chlorine atom bonded as well. To the right, a ball-and-stick model illustrates the 3D arrangement, featuring a large green sphere for the chlorine atom, smaller grey spheres for the carbon and hydrogen atoms, and sticks representing chemical bonds.

[Generated by AI]

Arenes

Arenes are aromatic hydrocarbons with the term aromatic referring to their distinctive aromas. They contain a phenyl functional group, which is a cyclic structure consisting of six carbon atoms and five hydrogen atoms: in other words, a benzene ring minus a hydrogen atom. **Figure 14** shows the compound phenol, C_6H_5OH , which has a hydroxyl group bonded to the phenyl functional group.

Higher level (HL)

The structure of benzene is covered in section S2.2.12

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/benzene-hl-id-45135/>).

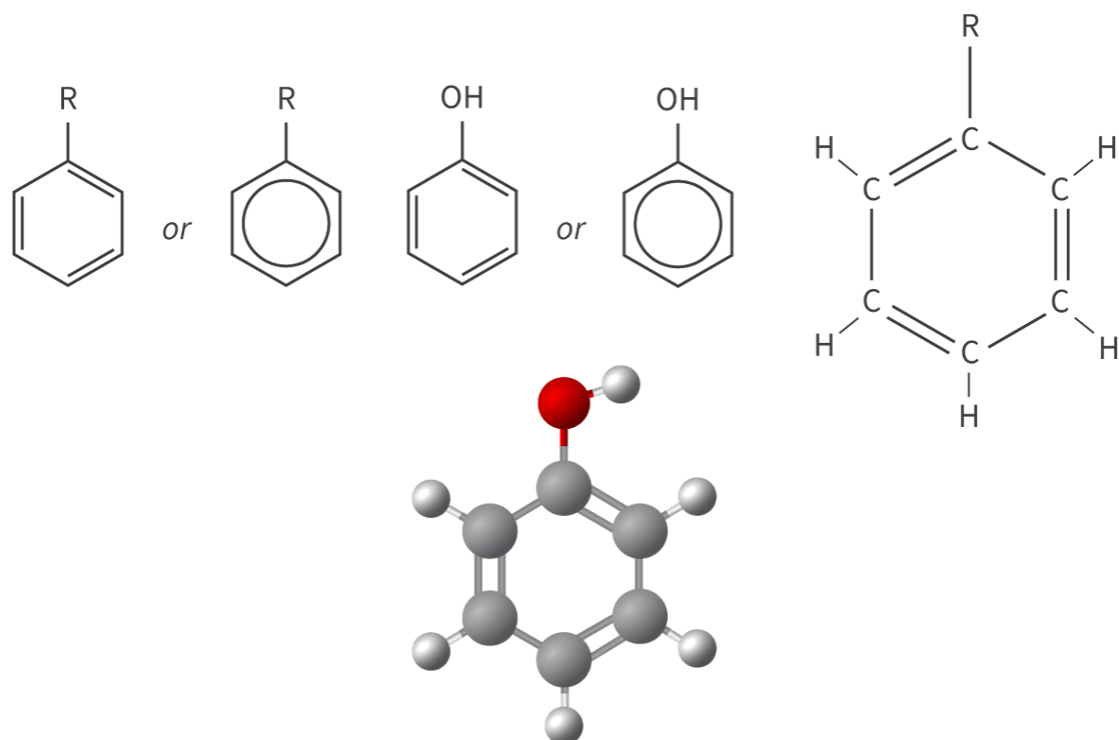


Figure 14. The phenyl functional group (left) and the full structural formula of phenol (right) and the ball and stick model (bottom).

 More information for figure 14

The image shows a chemical structure diagram of phenol.

1. Top Part: Displays two types of phenyl groups: one with a single hexagonal benzene ring labeled with alternating single and double bonds (R-connected) and another labeled with an overlaying circle, indicating delocalized pi electrons.
2. Middle Part: Shows the structural formula of phenol. It includes a benzene ring with an OH group attached, represented in a stick formula. The oxygen is labeled as connecting to a hydrogen atom to denote the hydroxyl group.
3. Bottom Part: Depicts a ball and stick model of phenol. The hexagonal ring is made up of gray balls representing carbon atoms, with white sticks (bonds) connecting them. A red ball represents the oxygen atom of the hydroxyl group, and the small white ball next to it depicts the hydrogen atom.

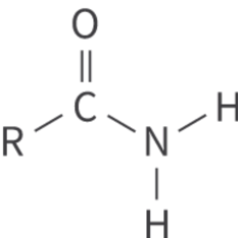
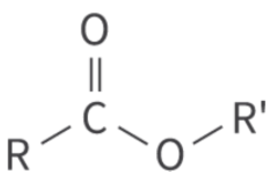
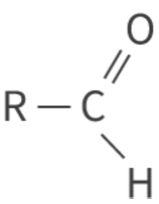
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Table 1 summarises the classes and functional groups that you need to learn.

Table 1. Classes and functional groups.

Class	Functional group name	Functional group structure	Example
Alkanes	N/A	$\text{—CH}_2\text{—CH}_2\text{—}$	H—
Alkenes	N/A	—CH=CH—	H—C=C—H H—C=C—H

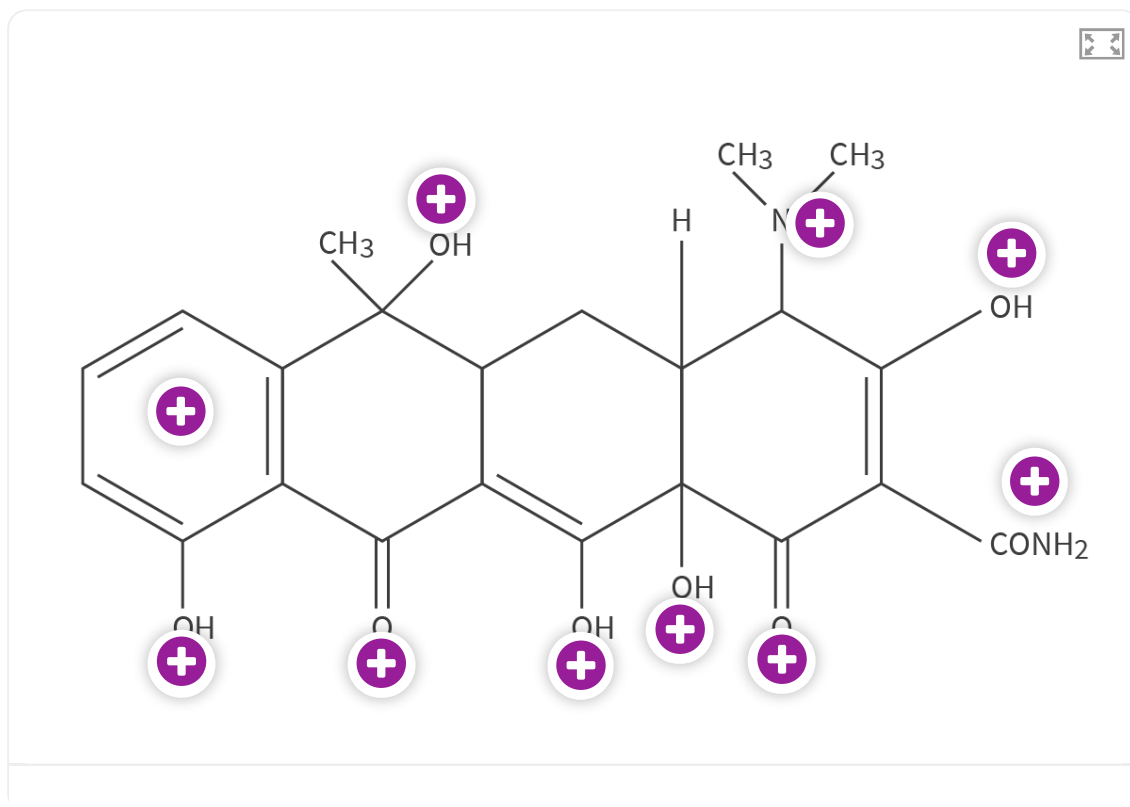
Alkyne	N/A	$-C \equiv C-$ 👁 More information	H-
Ethers	Alkoxy	$R-O-R'$ $R-O-R'$	Me H ₃ C
Amines	Amino	$R-NH_2$ $R-N \begin{matrix} H \\ \\ H \end{matrix}$ 👁 More information	N H

A m i d e s	A m i d o	$\text{R}-\text{CONH}_2$  More information	I (H₃C
Es te r s	Es te r	$\text{R}-\text{COO}-\text{R}' \text{ or } \text{R}-\text{CO}_2-\text{R}'$  More information	Metl H
Al d e h y d e s	C a r b o n y l	$\text{R}-\text{CHO}$  More information	H
K e t o n e s	C a r b o n y l	$\text{R}-\text{CO}-\text{R}'$	(

		$\begin{array}{c} \text{O} \\ \\ \text{R}-\text{C}-\text{R}' \end{array}$ 👁 More information	H ₃ C
C ar b ox yli c ac id s	C ar bo xy l	R—COOH or R—CO₂H $\begin{array}{c} \text{O} \\ \\ \text{R}-\text{C} \\ \\ \text{OH} \end{array}$ 👁 More information	Methyl
H al o g e n o al ka n es	H al og en o	R—X Where X is a halogen atom (F, Cl, Br or I) $\begin{array}{c} \text{H} \\ \\ \text{R}-\text{C}-\text{X} \\ \\ \text{H} \end{array}$ 👁 More information	Chloromethane

 More information

In **Interactive 1**, click the plus signs on the functional groups in the molecule shown. The name of the molecule is tetracycline, which is an antibiotic.



Interactive 1. Functional Groups.

 More information for interactive 1

An interactive chemical structure of tetracycline features four fused six-membered rings arranged in a straight line.

The first ring contains three delocalized pi electrons positioned at the top left, right, and bottom left. An OH group is single-bonded at the bottom.

The second ring features a double-bonded O at the bottom, along with a CH₃ group and an OH group single-bonded separately at the top.

The third ring contains one delocalized pi electron at the bottom left and an OH group single-bonded at the bottom.

The fourth ring contains one delocalized pi electron on the right and has an OH group single-bonded at the top right, a CONH₂ group single-bonded at the bottom right, an O double-bonded at the bottom, and an N single-bonded at the top. N is further single-bonded to two CH₃ groups.

An H atom is single-bonded at the fused area between the third and fourth rings at the top, and an OH atom is single-bonded to the fused area between the third and fourth rings.

Each functional group has a plus sign, indicating an interactive hotspot. Selecting a hotspot reveals the name of the corresponding functional group.

Hotspot on OH reads, Hydroxyl

Hotspot on double-bonded O reads, Carbonyl
Hotspot on CONH₂ reads, Amide
Hotspot on N (CH₃)₂ reads, Amino
Hotspot on the center of the first ring reads, Phenyl

This interactive structure helps users identify functional groups in tetracycline.

Worked example 1

The structure of the amino acid phenylglycine is shown. Which functional groups does it contain?

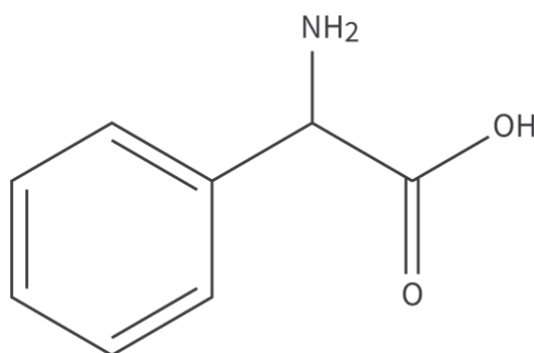


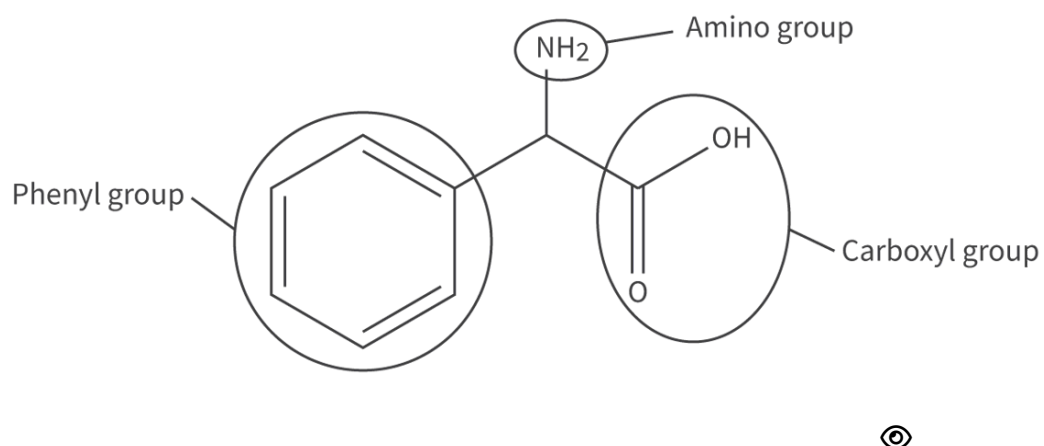
Figure 15. The structure of the amino acid phenylglycine.

 More information for figure 15

The image shows the chemical structure of the amino acid phenylglycine. It includes a benzene ring on the left with alternating double and single bonds, indicative of its aromatic nature. Attached to the benzene ring is a two-carbon side chain. The first carbon next to the benzene ring is connected to an amino group (NH₂) at the top. The second carbon is part of a carboxyl group (COOH) which includes a double-bonded oxygen (O) at the top and a hydroxyl group (OH) at the right. This structure highlights the functional groups present, including the aromatic benzene ring, the amino group, and the carboxyl group.

[Generated by AI]

The molecule contains an amino group, a phenyl group and a carboxyl functional group.



Practice dragging the image of each structure to match the name of the functional group in **Interactive 2**.

Interactive 2. Name the Functional Group.

 More information for interactive 2

An interactive table presents various functional group structures alongside empty labels. Users need to identify each functional group and drag and drop the correct group name into the corresponding fields.

The functional group structures are illustrated as follows:

1. R is single-bonded to O, which is single-bonded to R'.
2. R is single-bonded to N, which is single-bonded to two H.
3. R is single-bonded to C, which is double-bonded to O and single-bonded to another O. This second O is single-bonded to R'.

4. R is single-bonded to C, which is double-bonded to O and single-bonded to N.

This N is also single-bonded to two H.

5. R is single-bonded to C, which is double-bonded to O and single-bonded to H.

6. R is attached to a benzene ring.

7. R is single-bonded to C, which is double-bonded to O and single-bonded to R'.

8. R is single-bonded to C, which is double-bonded to O and single-bonded to OH.

9. R is single-bonded to OH.

10. R is single-bonded to C, which is single-bonded to two H and one X.

The drag-and-drop options provided are:

Amido

Amino

Carbonyl (aldehyde)

Carbonyl (ketone)

Carboxyl

Ester

Alkoxy

Halogeno

Hydroxyl

Phenyl

This interactivity helps users recognise and correctly name common functional groups based on their structural features.

Read below for the solution:

Structure 1: Alkoxy

Structure 2: Amino

Structure 3: Ester

Structure 4: Amido

Structure 5: Carbonyl (aldehyde)

Structure 6: Phenyl

Structure 7: Carbonyl (ketone)

Structure 8: Carboxyl

Structure 9: Hydroxyl

Structure 10: Halogeno

Aspect: Experiments

How can functional group reactivity be used to determine a reaction pathway between compounds, e.g. converting ethene into ethanoic acid (see subtopics R3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/print/>))

[753301/book/the-big-picture-id-43486/](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/)) and R3.4
[\(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43118/\)?](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43118/)

In this activity, you will practise identifying the functional groups covered in this section.

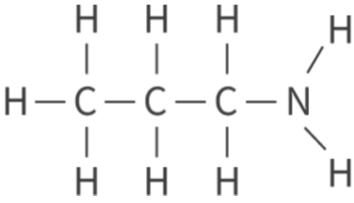
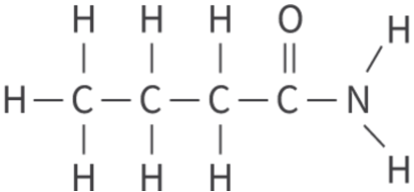
Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** individual activity

1. Identify the class and functional group from the full structural formulas shown.

	Compound
(a)	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{OH} \\ \quad \\ \text{H} \quad \text{H} \end{array} $ 👁 More information
(b)	$ \begin{array}{c} \text{H} \quad \quad \text{O} \\ \quad \quad // \\ \text{H}-\text{C}-\text{C} \\ \quad \quad \backslash \\ \text{H} \quad \quad \text{OH} \end{array} $ 👁 More information

	Compound
(c)	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad // \\ \text{H}-\text{C}-\text{C}-\text{C} \\ \quad \quad \backslash \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $ 👁 More information
(d)	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{Br} \quad \text{H} \quad \text{H} \end{array} $ 👁 More information
(e)	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \quad \\ \text{H} \quad \text{H} \quad \quad \text{H} \quad \text{H} \end{array} $ 👁 More information
(f)	$ \begin{array}{c} \text{H} \quad \text{O} \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \quad \quad \quad \\ \text{H} \quad \quad \quad \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} $ 👁 More information
(g)	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array} $

	Compound
	👁 More information
(h)	 <chem>CCCN</chem> 👁 More information
(i)	 <chem>CCCC(=O)N</chem> 👁 More information

2. Identify the functional group(s) from the condensed structural formulas.

- (a) $\text{CH}_3\text{CH}_2\text{COOH}$
- (b) HCHO
- (c) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$
- (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- (e) $\text{CH}_3\text{CH}_2\text{NH}_2$
- (f) $\text{C}_6\text{H}_5\text{CH}_3$
- (g) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$
- (h) CH_3CONH_2
- (i) $\text{CH}_3\text{CH}_2\text{Cl}$
- (j) $\text{CH}_3\text{COOCH}_2\text{CH}_3$

1.

	Class	
(a)	Alcohol	

	Class	
(b)	Carboxylic acid	
(c)	Aldehyde	
(d)	Halogenoalkane	
(e)	Ketone	
(f)	Ester	
(g)	Ether	
(h)	Amine	
(i)	Amide	

2. (a) Carboxyl
(b) Carbonyl (aldehyde)
(c) Carbonyl (ketone)
(d) Hydroxyl
(e) Amino
(f) Phenyl
(g) Ether
(h) Amido
(i) Halogenoalkane
(j) Ester